Modelling with Straight Lines

Exam code: 9MA0

Download PDF

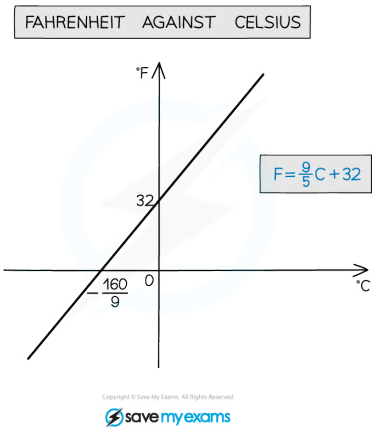
Written by: Paul
Updated on 20 June 2025

Did this video help you? Yes No

Modelling with straight lines

What is modelling with straight lines?

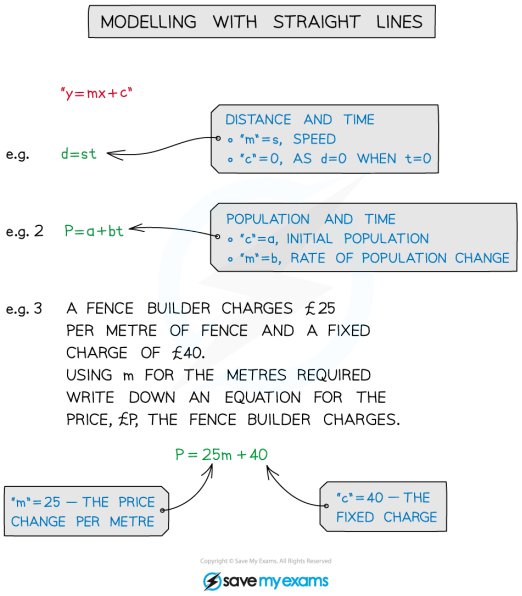
- **Modelling** means applying the mathematics to a **real-life** situation
- The model is used to **find** or **predict** future values
- Models will often need **refining** to improve their accuracy



Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- Turn the words from the question into mathematical equations
- In **y = mx + c** constants "**m**" and "**c**" will have a meaning
  - gradient "**m**" will be the **rate of change**
  - eg extension in a spring per 1 kg of mass added
  - **y**-axis intercept "**c**" will be the **initial value**
  - eg resting length of a spring



- The skills needed are covered in the other pages of the section Equation of a Straight Line

Worked Example

an initial population of 20. Scientists use the linear model $P = a + bt$ to predict the future population of the animal where P is the population and t is the time in years since the animal was first introduced to the area.

(a) Write down the value of the constant a.

(b) After 5 years scientists observe the population has tripled since the animal was first introduced to the area. Work out the value of b.

(c) Sketch the graph of population for the first 10 years

(d) Suggest one criticism of this model

P=a+bt

OF THE FORM "y=mx+c"

a) a=20

a IS THE INITIAL POPULATION, WHEN t=0

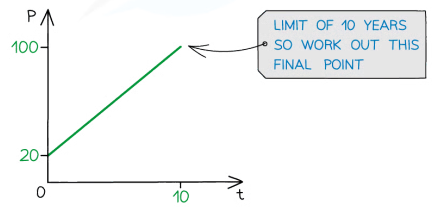
b) AT t=5, P=3×20=60

60=20+5b

b=8

a=20 FROM a)
P=60 WHEN t=5

c)



d) POPULATION INCREASES INDEFINITELY

POPULATION GROWTH IS USUALLY MODELLED EXPONENTIALLY — SLOW AT FIRST THEN MUCH QUICKER

Copyright © Save My Exams. All Rights Reserved
save my exams

Teacher only

New! Exclusive Teacher-Only Questions

Fresh, expert-written questions

Students can't access them

Fair accurate tests - built fast

Explore in Test Builder →

Test yourself

Was this revision note helpful? Yes No

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.